Sir Talha

SEAT NO. CT-005

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2021

Time: 3 Hours

Dated:13-02-2023
Max.Marks:60

Professional Ethics - HS-219

Q.1.

a) What is whistleblowing? What are the four conditions that make whistleblowing morally acceptable? [6-Marks] (CLO-1)

b) Define safety and risk. What are the factors that determine our assessment of risk in any given situation? [6-Marks] (CLO-1)

Q.2.

a) Why is bribery wrong? How would you determine whether something is a gift or a bribe? [6-Marks] (CLO-1)

b) What is a conflict of interest? What are the three types of conflict of interest? [6-Marks] (CLO-1)

Q.3.

a) Elaborate the limitations of human-centered ethics in relation to environmental ethics? Which environmental moral framework do you think is best suited to protect the environment, and why? [6-Marks] (CLO-1)

b) Explain what is 'rights ethics' and 'duty ethics'. What are the similarities and differences between these two moral theories? [6-Marks] (CLO-1)

Q.4. The intense pressure to get good grades in university often leads to temptations to cheat on exams or assignments. Cheating can take many forms, including copying someone else's work or using 'cheat sheets' during an exam.
Why is such cheating wrong? How would you formulate it as an ethical dilemma? Systematically argue against cheating using PEC codes of ethics, virtue ethics and another moral theory of your choice. [12-Marks] (CLO-2)

Q.5. The use of AI in judicial systems around the world is increasing, creating more ethical questions to explore. AI could presumably evaluate cases and apply justice in a better, faster, and more efficient way than a judge.
AI methods can potentially have a huge impact in a wide range of areas, from the legal professions and the judiciary to aiding the decision-making of legislative and administrative public bodies. For example, they can increase the efficiency and accuracy of lawyers in both counselling and litigation, with benefits to lawyers, their clients and society as a whole. Existing software systems for judges can be complemented and enhanced through AI tools to support them in drafting new decisions. This trend towards the ever-increasing use of autonomous systems has been described as the automatization of justice.
Some argue that AI could help create a fairer criminal judicial system, in which machines could evaluate and weigh relevant factors better than human, taking advantage of its speed and large data ingestion. AI would therefore make decisions based on informed decisions devoid of any bias and subjectivity.
Identify the potential ethical challenges and risks of the automatization of justice around the world. Moreover, would you want to be judged by a robot in a court of law? Would you, even if we are not sure how it reaches its conclusions? Provide your moral reasoning with reference to ethical theories, where possible. [12-Marks] (CLO-2)

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Sir Talha J. Sarhandi

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NED University of Engineering and Technology - Department of Computer Science and IT

Professional Ethics (HS-219) - Midterm Examination Fall 2022

CLO 1: Discuss the contemporary frameworks of professional and engineering ethics in the light of ethical theories and dilemmas. (C-2)

CLO 2: Apply principles, theories, and codes of ethics in situations related to professional practice. (C-3)

CLO 3: Value professional, aspirational, and collective ethics for continual professional development. (A-3)

MAX TIME: 90 Minutes **MAX MARKS: 20**

Q.1) What are the different ways to study morality in philosophy? Situate professional ethics within the philosophical study of morals. **[CLO-1] [4-Marks]**

Q.2) What is utilitarianism? Explain its features. Give an example of a moral decision you made in your student or professional life based on the utilitarian principle. **[CLO-1] [6-Marks]**

Q.3) You discover that a colleague has been downloading restricted files that contain trade secrets about a new product that the colleague is not personally involved with. You know the colleague has been having financial problems, and you fear the colleague is planning to sell the secrets or perhaps leave the company and use them in starting up his own company. Company policy requires you to inform his supervisor, but the colleague is a close friend. What should you do as a moral and responsible professional? **[CLO-2] [4-Marks]**

Q.4) Carefully read the case given below, and answer the following question. **[CLO-2] [6-Marks]**

U.S. consulting firm Arthur D. Little International Inc. studied the impact of smoking on Czech public finances at Philip Morris' request in November 2000. The study found the cost benefits of smokers' early mortality, together with cigarette-tax revenue, outweighed the economic drawbacks of health-care and other smoking-related costs.

The study was commissioned by the tobacco firm's Czech unit in response to claims by the Czech health ministry that smoking's costs outweighed its fiscal benefits. It found the impact of smoking on Czech public finance to be a net gain of about $147 million. Most of that gain was derived from tax revenue. Moreover, the government also saved between about $24 million to $30 million in health-care, pension and public-housing costs due to the early deaths of smokers -- one of the 'positive' benefits of smoking, according to the study.

Patti Lynn, associate campaign director for corporate watchdog group Infact, which has battled tobacco companies over their marketing practices expressed horror at the study and said, "Even if it were true that smokers dying young would save money for the economy, it's a real scary logic on which to base policy."

Do you agree or disagree with the kind of cost-benefit analysis provided in the case? Illustrate your opinion with adequate moral reasoning.

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SEAT NO. CT-21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2021

Time: 3 Hours

Dated:13-02-2023
Max.Marks:60

Professional Ethics - HS-219

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Identify the potential ethical challenges and risks of the automatization of justice around the world. Moreover, would you want to be judged by a robot in a court of law? Would you, even if we are not sure how it reaches its conclusions? Provide your moral reasoning with reference to ethical theories, where possible. [12-Marks] (CLO-2)

SEAT NO. CT-030

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2021
BATCH 2020

Dated:15-02-2022

Time: 3 Hours **Max.Marks:60**

Professional Ethics - HS-219

INSTRUCTIONS: 1) Attempt all the questions.
2) Start every answer on a fresh page.

Q1. Read the given case carefully and answer the questions that follow. [20] [CLO-2]

Jeremy is a fellow for his University's Global Service fellowship program. In his role, he must work with an international non-profit organization, SolarSolver, to develop a portable solar panel kit for rural villages in Belize – a country in Central America.

During a visit to SolarSolver's headquarters, one of the organization's representatives, Shiraine, asked Jeremy about using fellowship's grant money to purchase items not readily available in Belize for SolarSolver. Since fellowship funds were obtained through a reimbursement process, she wanted Jeremy to falsify receipts from solar-kit related purposes to obtain the money.

Jeremy's first instinct was that it would be unethical to use money intended for one purpose to fulfill another. However, he recalled Shiraine doing him several favors in the past. For instance, prior to his admittance into the fellowship, Jeremy had worked closely with Shiraine and other SolarSolver representatives to develop a proposal for his project. If it were not for her help (as well as her company's help in sponsoring the project), Jeremy strongly believed that he would not have received the fellowship.

Another time, a power outage occurred while he was demonstrating the solar-kit's operation and installment to local villagers in Belize. Jeremy tried calling SolarSolver's headquarters, but it was already past sunset and nobody was in the office. Finally, he decided to call Shiraine's house for help. To his relief, Shiraine answered and immediately sent a team with portable light generators to Jeremy's rescue. Jeremy is grateful for Shiraine's acts of kindness, but he does not believe that falsifying receipts is the right thing to do.

(a) Identify and state precisely the ethical dilemma faced by Jeremy in this case. [4]
(b) What is the unambiguously morally acceptable course of action (positive paradigm) for Jeremy in this case? [3]
(c) What is the unambiguously *not* morally acceptable course of action (negative paradigm) for Jeremy in this case? [3]
(d) What is your proposed solution (problem) for Jeremy? [2]
(e) Use the line drawing technique to answer "what should Jeremy do?" [8]

Q2 (a). Mention four of the myths about sexual harassment. [4] [CLO-1]

Q2 (b). Read the following case and identify the form/type of sexual harassment that is at work here. Then suggest what should Holly do. Should she say something? [6] [CLO-1]

Holly has been working at a large construction company for three years. Although an intern, Holly has earned the respect of her peers. Her supervisor (and project manager), along with her coworkers, constantly support her by teaching her new materials and encouraging her to tackle new tasks. For one specific project, her

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supervisor chose her to visit the job site for the construction of flood retaining walls. Holly is familiar with the protocol required by her company, and her boss trusts her to always wear protective equipment every time she goes into the field.

However, when she went to the job site for this project, she heard someone yell, "Be careful not to break a nail!" while she was walking down a particularly steep mud slope. She dismissed the comment at first, but throughout the remainder of the project, different construction workers would make condescending remarks and gestures at her. They would call her "Princess" or hold out their arms as though expecting her to fall, slip, or hurt herself.

Holly feels she is being treated disrespectfully because of her gender, but never brings it up to her supervisor as the workers were subcontracted from various companies by the General Contractor. Holly doesn't feel right getting other people in trouble for such little comments, and is worried long term about how her success might be affected if she makes a complaint.

Q3 (a). What are the four forms/types of whistleblowing? Define each. [4] [CLO-1]

Q3 (b). Read the following case and identify if the offer involved is a bribe or a gift. Explain why you think if it is a bribe or a gift. [6] [CLO-1]

Simon Williams works for Upscale Company. He has been assigned to deal with vendors who supply needed parts to the Upscale Company. Henry Bloom, sales representative from one of Upscale Company's regular vendors, plays in the same golf club as Simon. One evening they find themselves playing together. Sometime during the game Simon mentions that he is really looking forward to vacationing in Florida (a state in the USA) next month. Henry says his uncle owns an apartment in Florida that he rents out during the months he and his family are away vacationing. Henry offers to find out if the apartment is available next month. He assures Simon that the rent of the apartment would be quite low. This leaves Simon wondering if he is being offered a bribe or a gift. What do you think?

Q4 (a). Mention and define any four forms of academic dishonesty among students. [4] [CLO-1]

Q4 (b). James is a third-year computer science student who has been placed on probation for a low GPA, even though he knows he is doing the best work he can. A concerned friend, Susan, offers to help James by sitting next to him and "sharing" her answers during the next exam. What form of academic dishonesty is Susan engaging in? Briefly explain your answer. [6] [CLO-1]

Q5 (a). Briefly explain the concepts of "the invisible hand" and "the tragedy of the commons". [4] [CLO-1]

Q5 (b). A team of biomedical computer scientists develop a system for identifying people from a distance of up to 200 metres. A short tube attached to a sophisticated receiver and computer, and aimed at a person's head, reads the individual's unique pattern of brain waves when standard words are spoken. The team patents the invention and forms a company to manufacture and sell it. The device is an immediate success within the banking industry. It is used to secretly verify the identification of customers at tellers' windows in the bank. The computer scientists, however, refused to take any responsibility for such use of the device without customer notification or consent. They argue that the companies that buy the product are responsible for its use. They also refuse to be involved in notifying the public about the product's availability and the way it is being used. Does employing the device without customer awareness violate the right to privacy or to informed consent? Do the computer scientists perhaps have a moral obligation to market the product with suggested guidelines for its ethical use? [6] [CLO-1]

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SEAT NO. CT-030

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2021
BATCH 2020

Dated:15-02-2022

Time: 3 Hours **Max.Marks:60**

Professional Ethics - HS-219

INSTRUCTIONS: 1) Attempt all the questions.
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NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (COMPUTER SYSTEMS ENGINEERING)

SPRING SEMESTER EXAMINATIONS 2020

BATCH 2018

Dated:25-09-2020

Time: 2 Hours

Max.Marks:30

Professional Ethics - HS-219

Instructions

Attempt all questions. Be relevant and legible in your answers. Marks and CLO for each question are given at the end of the question.

Q1. How can risk apprehensiveness/fear be reduced? Discuss with the reference to any three factors. [5] CLO1

Q2. Read the given case study carefully and answer the following questions. [8] CLO2

"Engineer A was Chief Engineer of a plant that processed raw ore. The refining process involved several dangerous chemicals, which were re-captured and re-circulated; however, careful operation was essential to prevent these chemicals from escaping into the wastewater. Engineer A worked alongside the Operations Manager, and both of them reported to the Plant Director. Engineer A was responsible for technical matters, such as design, maintenance, and safety. The Operations Manager was responsible for hiring, scheduling, and meeting production targets. Both the Operations Manager and the Plant Director were older than Engineer A, but neither was a Professional Engineer nor a Professional Geoscientist.

During the first few months on the job, Engineer A reviewed, updated, and improved the plant *Operating Manual* prepared by the previous Chief Engineer. Engineer A ensured that copies of the manual were available to the plant operating staff and personally conducted several training sessions for key operating staff. In spite of these efforts, however, Engineer A observed many infractions of the *Operating Manual* throughout the plant, and he could see that the toxic chemicals were possibly escaping into the wastewater. Engineer A considered this lax attitude toward safety to be very risky. Tests of the wastewater effluent showed wide variations of the escaping chemicals, with concentrations that occasionally reached the legal limits. On several occasions, Engineer A initiated disciplinary measures against operating staff, but these were dealt with lightly by the Operations Manager, for whom the staff worked. Engineer A eventually came to understand that the Operations Manager put production ahead of safety and was casual

about enforcing the safety provisions in the *Operating Manual*.
Finally, Engineer A warned the Operations Manager about these unsafe practices in writing and demanded that infractions be disciplined more severely. As a last resort, Engineer A went

directly to the Plant Director and explained the problem, but the Director simply said, “Work it out among yourselves.”

Question: *If you were Engineer A, what would you do at this point?”* support your answer with ethical theories PEC and IEEE Code. Your answer should be in continuous writing paragraph form.

Q3. Suppose you are head of a department, someone brings you a gift but you are not sure whether it is a gift or bribe. What three strategies will you use to ensure that the brought item is a gift, briefly explain each strategy? [6] CLO 3

Q4. Conflicts are inevitable in the workplace. Suggest three principles to manage conflict, discuss each principle concisely. [5] CLO 4

Q5. Collective bargain (union) has both merits and demerits. Discuss both sides of the collective bargain and support each side with two points. [6] CLO 4

NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY
SCIT (COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
FALL SEMESTER-2021
BATCH 2020-2021
MIDTERM EXAMINATION

PROFESSIONAL ETHICS (HS-219)

Date: 14-12-2021

Time: 2 hours
Max. marks: 20

INSTRUCTIONS: 1) Attempt both the questions.

Q1. Rachel works as a Quality Assurance Engineer at a large electronics company. She is responsible for the final testing of her company's servers and is part of a team which decides when new products will be shipped to distributors for sale.

Rachel's company has a contract with another company which makes the chips which are incorporated into the servers Rachel's company makes. The business model for this product is to release a new generation server approximately every six months, meaning Rachel has a limited timeframe to conduct her Quality Control tests.

Because there is such a short amount of time between the release of each next new product, the Quality Assurance department cannot perform every possible test on the servers to ensure they are defect free. Rachel will not ship a product if there is any possibility that the server could malfunction and cause physical harm to the customer. However, she will ship a product that has a higher likelihood of failure resulting in data loss for the customer, because she knows that if she doesn't, her company's competitor will. Is this an ethical way to conduct business? What would utilitarians, duty ethicists, and rights ethicists think of this way of doing business? [08] (CLO-1)

Q2. Write a detailed note on the ethics of duty explaining the three principles upon which is founded Immanuel Kant's ethics. [12] (CLO-1)